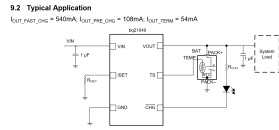


Calcs:
<https://docs.google.com/spreadsheets/d/1r5jx09jgPo7Qa4dIGZ32ZsXNAvsbSHH8eOILfTHGSeW/edit?usp=sharing>

$R_{ISET} = [540\text{A}\Omega / 0.45\text{A}] = 1.2\text{ k}\Omega$

To Disable the temp sense function, use a fixed 10k resistor between the TS (terminal 1) and Vss.

LED Status: connect a 1.5k Ω resistor in series with a LED between the OUT terminal and the CHG terminal.



Battery Life

Battery: 500mAh, 3.7V

Discharge

Running

500mAh/20mA = 25h

$500\text{mAh} / (30\text{mA} * 6.4\text{V} / 3.7\text{V} / 0.8 + 1.65\text{mA} + 2.1\text{mA}) = 7.2\text{h}$
 $(30\text{mA} * 6.4\text{V}) / 0.8 / 3.7\text{V} = 65\text{mA} + 1.65\text{mA} + 2.1\text{mA} = 68.8\text{mA}$

20mA = 47mA = 10.6h

LED driver quiescent current = 1.65mA max

LED driver efficiency (5% duty) = 80%

MCU quiescent current: 2.1mA

Shutdown

$500\text{mAh} / (.0071) = 70,422\text{h} = 8\text{y}$

MCU shutdown current: 6.1uA

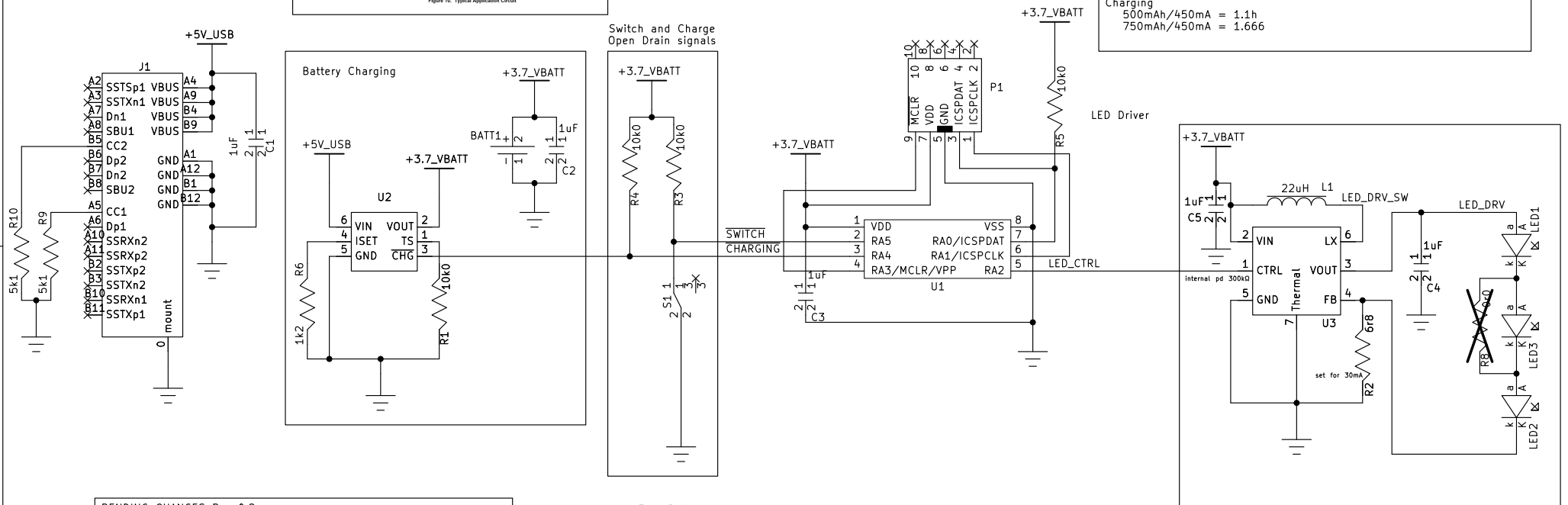
LED Driver Shutdown current: 1uA

cap leakage, etc.: ?

Charging

$500\text{mAh} / 450\text{mA} = 1.1\text{h}$

$750\text{mAh} / 450\text{mA} = 1.66\text{h}$



PENDING CHANGES Rev 0.2

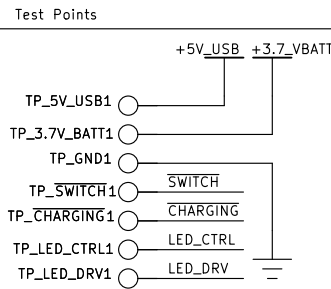
[X]—Measure Battery Voltage? — move measurement to different pin

[X]—Add another LED

[X]—Battery Connector

[X]—USBC — 5k1 R from CC to GND

[X]—0402 components for more space, smaller MCU



TGMF LLC

Sheet: /

File: salt_booklamp.kicad_sch

Title: SaltGlow

Size: A4 Date: 2025-06-23

KiCad E.D.A. 10.0.0

Rev: 0.2

Id: 1/1